

# COMIC CRAFT AI

Team leader: Vinoth kumar.R

NMID: 50E47D54BFE7D20A8C74ACAC3473F525

Team members:

1.Barath Sriram .J

NMID: 747DBCC7B1E819189634C0B43276DFCE

2.Deepan raj.A

NMID: 1CB77598E337BF18A4E4BC6633653678

3.Niranjan.M

NMID: E69C456C5E1A3418984471C294E22645

# ComicCraft

## *AI Comic Story Creator using Gemini Models*

### Project Description

**ComicCraft** is a web-based application that uses AI to generate personalized comic book stories and illustrations based on user-provided prompts. Built with FastAPI and integrated with Google's Gemini AI models along with Stable Diffusion, ComicCraft streamlines the creative process of generating storylines, dialogues, and vivid comic-style imagery automatically.

The application takes in user information—such as story prompt, main character name, setting, tone, and art style—and generates a panel-by-panel storyline and corresponding illustrations. Users can view the comic preview directly on the web interface and also download the full comic in PDF format, structured with both images and narration.

With a clean, interactive HTML interface and efficient image generation powered by Hugging Face's Diffusers library, ComicCraft ensures that each story feels cohesive and visually engaging. It offers both API endpoints and an intuitive frontend for ease of use, making comic creation accessible even to non-artists and casual users.

## Scenarios

### Scenario 1: Personalized Comic Generation

A user enters a prompt describing their comic story, such as “A brave fox exploring an enchanted forest.” The system uses this input to generate a complete comic strip, including multiple panels with accompanying images and text. The user also specifies preferences like the main character's name, the setting (e.g., forest), tone (e.g., dramatic), and art style (e.g., anime). The system combines Gemini 1.5 Pro for story generation and Stable Diffusion for image creation to produce a visually immersive, personalized comic strip.

### Scenario 2: Mood & Style Customization

A user wants a light-hearted, cartoonish feel for their comic. On the input form, they select “funny” as the tone and “comic book” as the art style. When they submit, the system regenerates the entire pipeline—outline, story, and images—this time tailored to produce a more humorous narrative and a classic comic-book look. This lets users iterate on and customize the mood and visual style of their comic until it matches their creative vision

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### Scenario 3: Export to PDF

After reviewing the on-screen preview of their comic, the user decides to download it. The system then:

- **Layout Binding:** Uses `layout_builder.py` to assemble each panel's title, image, and text into a structured layout.
- **PDF Generation:** Calls `exporters.py` to compile the comic into a PDF file using FPDF, placing images and narrative text onto consecutive pages and saving the PDF with a timestamped filename.
- **Export Success:** The user is redirected to an export success page, confirming the successful download. This feature ensures users can save, print, or share their comic in a professional PDF format.

Technical Architecture

The architecture of ComicCraft is structured into three primary components:

- **Frontend** (HTML, CSS, Jinja2) — captures user inputs and displays generated comics.
- **Backend** (FastAPI application) — handles data processing and routing.
- **AI Integration** (Google Gemini APIs and Hugging Face Diffusers) — generates story content and illustrations.
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Pipeline Flow

User (Browser)

- FastAPI Backend (app/main.py)
- HTML Templating (Jinja2 + CSS)
- index.html (Input Form) | AI Comic Generation Logic (routes.py + Gemini + Diffusers)
- Gemini Flash → Comic Outline
- Gemini Pro → Full Story Narration
- Diffusers (Stable Diffusion) → Panel-wise Image Generation
- Layout Builder (layout\_builder.py)
- Comic Exporter (exporters.py → PDF via FPDF)
- comic\_preview.html (Comic Preview Page)
- export\_success.html (Download Confirmation Page)

Selected AI Models

| Model                                             | Role                                                                                |
|---------------------------------------------------|-------------------------------------------------------------------------------------|
| Gemini Flash (models/gemini-1.5-flash)            | Fast, structured generation of the 5-panel comic outline based on the story prompt. |
| Gemini Pro (models/gemini-1.5-pro)                | Detailed, creative comic narration and character dialogue from the outline.         |
| Stable Diffusion (runwayml/stable-diffusion-v1-5) | Generates high-quality comic-style illustrations from image prompts.                |

Project Structure

```
comiccraft/
├── app/
│   ├── main.py
│   ├── routes.py
│   ├── gemini_flash.py
│   ├── gemini_pro.py
│   ├── image_generator.py
│   ├── layout_builder.py
│   └── exporters.py
├── templates/
│   ├── index.html
│   ├── comic_preview.html
│   └── export_success.html
├── static/
│   ├── panels/      (generated panel images)
│   ├── exports/     (exported comic PDFs)
│   └── fonts/        (fonts for PDF generation)
├── .env              (environment variables / API keys)
└── requirements.txt  (Python dependencies list)
```

Core Backend Functions

| Function           | File            | Purpose                                                                                                                                        |
|--------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| generate_outline() | gemini_flash.py | Uses Gemini Flash to generate a structured 5-panel comic outline (panel number, title, scene description, image prompt) from the story prompt. |
| generate_story()   | gemini_pro.py   | Uses Gemini Pro to expand the panel outline into a full comic-style story with narration and character dialogue.                               |

| Function             | File               | Purpose                                                                                                                       |
|----------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| generate_image()     | image_generator.py | Uses Stable Diffusion to create a comic-style image from the image prompt; sanitizes the filename and saves to static/panels. |
| build_comic_layout() | layout_builder.py  | Organizes generated images and story text into a structured layout, matching each image to its panel.                         |
| save_pdf()           | exporters.py       | Compiles the full comic into a multi-page PDF using FPDF and saves it with a timestamped filename to static/exports.          |

### API Routes

| Route                | Description                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| /                    | Loads the homepage (index.html) where users submit their story details.                                                        |
| /generate            | Handles form submission, processes the input using AI, generates the comic panels, and returns the comic preview page.         |
| /generate-comic/json | An API route that accepts JSON payloads, triggers comic generation, and returns comic layout data plus the generated PDF path. |
| /export-success      | Displays a success confirmation page after the comic is downloaded.                                                            |
| /test-image          | A developer utility route to test image generation from a direct prompt without full comic creation.                           |

## Frontend Pages

- **index.html:** A user-friendly form where users input the story prompt, character name, setting, tone, and preferred art style. Submits data to the /generate route via POST.
- **comic\_preview.html:** Displays the AI-generated comic panel-by-panel — panel title, scene illustration, scene description (italics), captions and narration text.
- **export\_success.html:** Displays a success message after the user downloads their comic as a PDF, with a call-to-action to create another comic.
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## Project Workflow / Milestones

### Milestone 1: Model Selection & Architecture

Research and select the generative AI models; define the application architecture; set up the development environment.

### Milestone 2: Core Functionalities Development

Develop the core functionalities; implement the FastAPI backend to manage routing and user input processing.

### Milestone 3: routes.py Development

Write the main application logic in routes.py, wiring each route to its backend AI workflow.

### Milestone 4: Frontend Development

Design and develop the user interface; create dynamic templates with FastAPI's Jinja2.

### Milestone 5: Deployment

Prepare the application for local deployment; test and verify local deployment.

## Pre-requisites

- FastAPI Framework Knowledge
- Uvicorn ASGI Server
- Python Programming Proficiency
- HTML, CSS, and Jinja2 Template Skills
- Google Gemini API Familiarity
- Hugging Face Diffusers & Stable Diffusion
- PyTorch Basics
- PDF Generation with FPDF
- Version Control with Git
- Python Package & Environment Setup (pip, virtualenv, or conda)

## Setup Commands

```
python -m venv comiccraft-env
```

```
comiccraft-env\Scripts\activate # Windows
```

```
pip install fastapi uvicorn jinja2 python-multipart google-generativeai diffusers transformers fpdf
```

```
Pillow accelerate
```

```
uvicorn app.main:app --reload
```

Visit <http://127.0.0.1:8000> for the app, and /docs for interactive API testing.

Project output:

01 IDEA →

02 STORY →

03 ART →

04 COMIC

## 1. STORY PROMPT

Describe your story concept or plot

Example: A brave fox exploring an enchanted forest

Please fill out this field.

## 2. MAIN CHARACTER NAME

Hero or protagonist

Enter your hero's name

## 3. SETTING

🌲 Forest



## 4. STORY TONE

💡 Light-hearted



## 5. ART STYLE

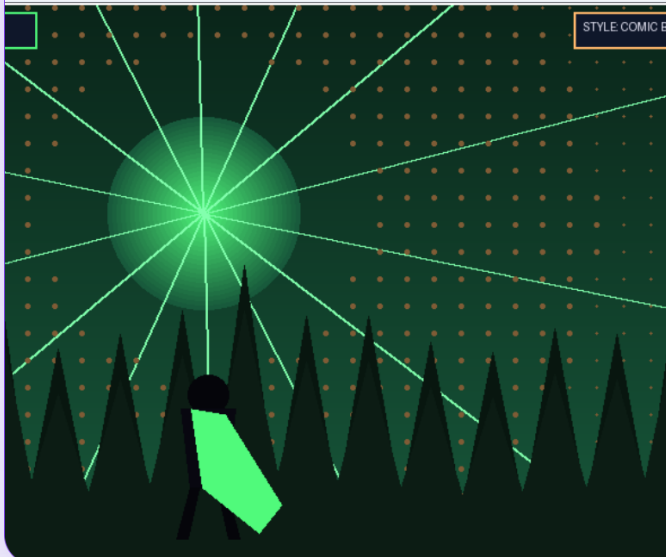
🌟 Comic Book



GENERATE COMIC

PANEL #01

A WORLD AWAKES



INTO THE HEART OF FOREST...

## STORY NARRATION

The tranquil silence of Forest was broken by the quiet footsteps of arjun, driven by the whispers of an untold destiny.

## ARJUN / DIALOGUE

arjun: "There is no turning back now. The answer lies somewhere deep within these grounds."

## SCENE DESCRIPTION

In the vibrant expanse of Forest, arjun stands at the threshold of a new journey. The atmosphere is thick with light-hearted energy as signs of a greater mystery begin to unfold.

[View AI Image Prompt](#)

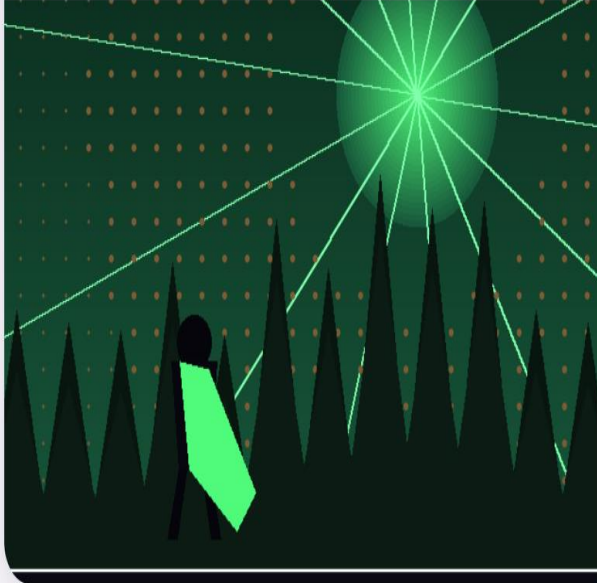


**COMICCRAFT**

AI COMIC STORY CREATOR

Dark Mode

[← Create Another Comic](#)



*As dawn broke over Forest, the challenge had been overcome, leaving arjun ready for whatever adventures lay beyond.*

**ARJUN / DIALOGUE**

**arjun:** "We did it. And this is only the beginning of our story."

**SCENE DESCRIPTION**

With peace restored to Forest, arjun gazes toward the sprawling horizon. The legend of this adventure has only just begun.

[▶ View AI Image Prompt](#)

[← CREATE ANOTHER COMIC](#)

[↓ DOWNLOAD YOUR COMIC AS PDF](#)



## ***YOUR COMIC IS READY!***

Your AI-powered comic has been successfully created and downloaded.

Open your downloaded PDF to view the printable edition with full artwork, captions, and narrative panels.

File: `comic_arjun_20260926_125326_101d.pdf`

↓ **DOWNLOAD AGAIN**



**CREATE ANOTHER COMIC**

GitHub Link: <https://github.com/kumarvinoth94332-jpg/comic-craft.git>

Project Link: The app is available at <http://127.0.0.1:8000>

🔗 Open : <http://127.0.0.1:8000>